Session 7: Solutions and facilitation notes

Answer keys for `Course/Practicals/s7_exercises.md` (Session 7 exercise sheet), Parts A–C. Numbers in Part C are from the results pack of each question (`Course/Data/results_packs/QN/results_pack_QN.md`, SIMULATED data), and each one names its table in the pack. These keys are for the instructor.

Part A: Integrity scenarios: green, amber or red?

#	Colour	Why?
1	**RED**: fabrication	Values "remembered" are invented data.
2	**RED**: falsification	Removing data after seeing that they weaken the result, without saying so, changes the findings.
3	**RED**: gift authorship	Seniority is not a contribution. All four ICMJE criteria are needed.
4a	**GREEN** (with disclosure)	Language editing of the author's own text, checked by the author and disclosed, is accepted by ICMJE and COPE. The author remains responsible.
4b	**RED**: breach of confidentiality	Identifiable patient data must never be pasted into a public AI tool. The analysis is also not accountable or reproducible.
5	**AMBER to RED**: salami slicing or redundant publication	Splitting one cohort into undisclosed separate papers distorts the evidence and can double-count patients.
6	**RED** (amber if minor and fixed before submission): plagiarism	Copying words verbatim without quotation marks is plagiarism even with a citation.

Instructor note: what to do instead (say it after each vote).

1. Record the values as missing and report how many are missing. Next time, use a data-collection timetable and check against source documents.
2. Pre-specify rules for implausible values in the data-quality plan (Session 5). Check the values against the source. If an exclusion is justified, report it and show the results with and without these patients.
3. Thank the head of department in the Acknowledgements, or invite a real contribution so that all four criteria are met. Agree the author list in writing at the protocol stage.
4. 4a: check the journal's AI policy and disclose the tool, version and purpose. 4b: de-identify the data, use institution-approved software, and have a named person run and check the analysis.
5. Publish one main paper with all pre-specified outcomes. Any genuine secondary paper must cite the main one and add something new.
6. Paraphrase in your own words and cite, or quote with quotation marks. Similarity-check software will flag copied text.

Instructor note: the example on the slide (a colleague lent an ultrasound machine and asks for authorship) is **AMBER/RED**. Providing equipment is not authorship, so thank them in the Acknowledgements.

Instructor note: facilitation. Run each scenario as a three-option poll. When the poll is split (often on 4a and 5), ask one person from each side for a reason in the chat, then state the rule. Honest error is not misconduct, but it must be corrected openly.

Part B: Proposal builder 7: model answers for the running example

Running example: a retrospective cohort of fluid balance and 28-day mortality in adults with septic shock in our ICU. Fill in each block live on the shared screen, then participants write their own.

B1. Ethics and consent

Question	Model answer
Which ethics committee, and what type of review?	The institutional research ethics committee, before any data collection. We expect expedited review because the data come from routine charts with no change in care. The committee decides.
Who are the participants, and can they decide for themselves?	Adults with septic shock admitted to our ICU. Most cannot decide for themselves when they become eligible (shock, sedation, ventilation).
Who gives consent?	We request a waiver of individual consent. The study is retrospective, carries no more than minimal risk, and many patients lack capacity or have left hospital. If the committee requires consent, the patient gives it when able, otherwise the legal representative.
Who takes consent, when and where?	No one, if the waiver is granted. If consent is required, a trained study nurse who is not the treating clinician asks the patient once able to decide, or the representative at visiting time, in the ICU.
Risks and burdens, and how they are minimised	No change in care, so no physical risk or extra burden. The main risk is loss of confidentiality. Coding, a separate identifying file and restricted access reduce it (B2).
Does the study need registration?	Not mandatory for an observational study. Recommended, for example on ClinicalTrials.gov before data extraction starts.

B2. Confidentiality

Question	Model answer
Which direct identifiers, and why?	Name, hospital number and date of birth, only to find the charts and check data against the source. They go on the identifying file only.
How will each patient get a study code?	A sequential code (FLU-001, FLU-002 and so on) given when the patient is entered.
Where is the identifying file kept, and who can open it?	On the institution's password-protected drive, separate from the data. Only the principal investigator and one data manager can open it.

Question	Model answer
Where will the de-identified data be stored?	On the institution-approved, password-protected server. The analysis file holds the study code, age, admission month and the study variables only.

B3. Feasibility check

☐	Question	Model comment
☐	**Time**	Ethics 1 to 3 months, data collection 12 months, then analysis and writing.
☐	**Money**	Staff time for data entry, printing, open-access fee.
☐	**People**	Two nurses per shift trained on the CRF. Data checked weekly. A statistician does the analysis. Protected time agreed with the head of department.
☐	**Patients**	About 180 adults with septic shock last year. We expect 120 eligible patients in 12 months.
☐	**Data**	Fluid charts, SOFA and lactate are recorded routinely. Fluid output at night and at weekends is often incomplete.

Biggest feasibility risk: incomplete night and weekend fluid charts. One fix: a weekly check of a sample of charts against the source by the data manager.

B4. Publication plan

Item	Model answer
Working title	"Cumulative fluid balance and 28-day mortality in adults with septic shock: a retrospective cohort study"
Target journal (and a back-up)	A critical-care or critical-care-nursing journal indexed in MEDLINE (checked in the NLM Catalog), with APC and licence known. Back-up: a general critical-care open-access journal listed in DOAJ
Reporting guideline	STROBE (cohort), not CONSORT
Registration needed? Where?	Not mandatory for an observational study. Recommended, for example ClinicalTrials.gov (from B1)
Authors and their roles (CRediT)	PI: Conceptualization, Methodology, Writing (original draft). Two nurses: Investigation, Data curation. A third team member: Investigation, Writing (review and editing). Statistician: Formal analysis. Head of department: Supervision, only if all four ICMJE criteria are met (otherwise Acknowledgements)
Conflicts of interest and funding	None declared. Internal departmental support
Data-sharing statement	De-identified dataset and data dictionary available from the corresponding author under a data-use agreement, if the ethics committee and consent allow

Item	Model answer
Conference	National critical-care congress abstract after the first analysis

Structured-abstract skeleton

Section	Model line
Background	A positive fluid balance may be linked to death in septic shock, but local data are lacking. We aimed to assess its association with 28-day mortality.
Methods	Retrospective cohort of adults with septic shock in one ICU over 12 months. Exposure: cumulative fluid balance at day 3. Outcome: 28-day mortality. Logistic regression adjusted for severity.
Results (expected)	"n patients. 28-day mortality x% vs y%. Adjusted OR (95% CI)."
Conclusion	If positive: "A positive day-3 fluid balance was associated with higher 28-day mortality." If not: "We found no evidence of an association. The CI does not exclude a clinically relevant effect."

Instructor note: common errors to correct. "Consent will be obtained" with no who, when or how. Assuming a retrospective study needs no ethics review. Recruitment numbers copied from the sample-size calculation instead of the unit's admissions. No named place for the identifying file. CONSORT named for a cohort. "Impact" or "effect" in the title. An author list that includes people who only collected data. Expected results written as numbers ("mortality fell by 30%").

Part C: Manuscript workshop 1: the Results section for each question

Answers depend on the question each participant chose. All numbers are from that question's results pack (`Course/Data/results_packs/QN/results_pack_QN.md`, **SIMULATED DATA, NOT REAL PATIENTS**), and each one names its table. The model sentences on the sheet come from the Q1 model below. For Q2 to Q5, the tables list what a correct draft contains in each block.

Q1: model Results (~400 words)

Working title. Positive fluid balance by day 3 and 28-day mortality in sepsis: a multicentre cohort study.

Participants (block 1, STROBE 13, pack section 2). Of 620 patients enrolled, 2 had no day-3 fluid balance (weight or output missing), and 618 were included in the analysis. 527 had complete data for the adjusted models (91 without lactate or age).

Descriptive data (block 2, STROBE 14, Table 1). Mean age was 62.6 years (SD 15.2), 375 of 618 (61%) were men, and the median SOFA score was 7 (IQR 5 to 9). A positive balance by day 3 occurred in 373 of 618 patients. These patients were sicker at baseline: median SOFA 8 (6 to 10) vs 6 (4 to 8), vasopressor 205 (55%) vs 61 (25%) and mechanical ventilation 216 (58%) vs 97 (40%). They less often had heart failure, 43 (12%) vs 54 (22%). Lactate was missing in 91 of 618 patients, APACHE II score in 6, mean arterial pressure in 4 and age in 1.

Outcome data (block 3, STROBE 15, Table 2, Figure 1). By day 28, 134 of 373 patients (35.9%) with a positive balance and 45 of 245 (18.4%) with a negative balance had died (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4, chi-square p < 0.001). Kaplan-Meier survival at day 28 was 64.1% and 81.6% (log-rank p < 0.001, Figure 1).

Main results (block 4, STROBE 16, Tables 3 and 4, Figures 2 and 3). The crude odds ratio (OR) for death with a positive balance was 2.49 (95% CI 1.70 to 3.70, n = 618). After adjustment for age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, it was 1.62 (95% CI 1.02 to 2.59, p = 0.044, n = 527) (Table 3, Figure 3). The adjusted ORs for all covariates are shown in Table 3 and Figure 2. In the Cox model, the crude hazard ratio (HR) was 2.17 (95% CI 1.55 to 3.05, 179 deaths) and the adjusted HR 1.46 (95% CI 1.00 to 2.13, p = 0.053, 161 deaths) (Table 4). The global test of proportional hazards gave p = 0.10.

Other analyses (block 5, STROBE 17, Tables 5 and 6, Figure 4). In the 527 complete cases, the crude OR was 2.22 (95% CI 1.48 to 3.39). With multiple imputation (n = 620), the adjusted OR was 1.70 (95% CI 1.09 to 2.64) (Table 6). The association differed by heart failure (p for interaction = 0.031). The adjusted OR was 4.39 (95% CI 1.65 to 12.38) in patients with heart failure and 1.29 (95% CI 0.77 to 2.19) in those without (Table 5, Figure 4).

AI use. None.

Instructor note. Percentages from Table 1 are given as the pack rounds them. The 373 of 618 in the descriptive paragraph comes from the column headers of Table 1. The risk ratio in Table 2 (1.96) has no 95% CI in the pack, so the model reports the risk difference instead.

Q2: acute kidney injury (AKI)

Block	What a correct draft contains (pack table)
1	620 enrolled, 2 without a day-3 balance, 618 analysed, 527 in the adjusted models (section 2)
2	The positive-balance group was sicker: median SOFA 8 (6 to 10) vs 6 (4 to 8), vasopressor 205 of 373 (55%) vs 61 of 245 (25%). Lactate missing in 91 of 618 (Table 1)
3	AKI in 230 of 373 (61.7%) with a positive balance vs 71 of 245 (29.0%) with a negative balance (section 4). KDIGO (Kidney Disease: Improving Global Outcomes) stage 3: 108 (29%) vs 18 (7%) (Table 2). Any AKI by balance category: 71 of 245 (29%), 90 of 166 (54%), 82 of 130 (63%) and 58 of 77 (75%), with the percentages for AKI stage 2 to 3 and renal replacement therapy (RRT) (Table 3, Figure 1)
4	Per 10 ml/kg: crude OR 1.37 (95% CI 1.27 to 1.47), n = 618. Adjusted OR 1.27 (95% CI 1.16 to 1.40), p < 0.001, n = 527 (Table 4, Figure 2)
5	Positive vs negative balance: crude OR 3.94 (95% CI 2.80 to 5.60), adjusted OR 2.70 (95% CI 1.76 to 4.15). AKI stage 2 to 3: adjusted OR 1.23 per 10 ml/kg (95% CI 1.12 to 1.35) (Table 4)

Q3: ventilator-free days (VFD)

Block	What a correct draft contains (pack table)
1	620 enrolled, 2 without a day-3 balance, 618 analysed, 527 in the adjusted models. 313 were ventilated on admission (section 2)
2	The positive-balance group was sicker: median SOFA 8 (6 to 10) vs 6 (4 to 8), mechanical ventilation 216 of 373 (58%) vs 97 of 245 (40%). Lactate missing in 91 of 618 (Table 1)
3	Median VFD 20 (IQR 0 to 28) with a positive balance vs 28 (22 to 28) with a negative balance. Mean 15.2 (SD 12.4) vs 21.6 (10.5). VFD of 0: 137 (37%) vs 45 (18%). VFD of 28: 117 (31%) vs 133 (54%). Ventilated patients: median 12 (0 to 22) vs 23 (0 to 25) (Table 2, Figures 1 and 2)
4	Per 10 ml/kg: crude difference −1.48 days (95% CI −1.84 to −1.13), n = 618. Adjusted −0.61 days (95% CI −1.00 to −0.21), p = 0.003, n = 527 (Table 3)
5	Positive vs negative balance, adjusted: −2.78 days (95% CI −4.76 to −0.80), p = 0.006. Hodges-Lehmann difference −3.0 days (95% CI −5.0 to 0.0), Mann-Whitney p < 0.001 (Table 3)

Q4: admission lactate and missing data

Block	What a correct draft contains (pack table)
1	All 620 patients analysed. Lactate missing in 93 (15.0%). Complete-case models 527, multiple imputation 620 (section 2)
2	441 survived and 179 died by day 28. Non-survivors were older (mean 66.0 vs 61.2 years) and more often received vasopressors (114 [64%] vs 153 [35%]). Median lactate 3.1 (IQR 2.2 to 4.5) vs 2.3 (1.6 to 3.4) mmol/L (Table 1). Patients without a lactate were less sick: median SOFA 5 (4 to 6) vs 8 (6 to 10), vasopressor 14 (15%) vs 253 (48%), died 18 of 93 (19.4%) vs 161 of 527 (30.6%) (Table 2, Figure 1)
3	28-day mortality by lactate: below 2 mmol/L 32 of 172 (19%), 2 to 4 mmol/L 76 of 248 (31%), above 4 mmol/L 53 of 107 (50%), not measured 18 of 93 (19%) (Table 3, Figure 2)
4	OR per 1 mmol/L, unadjusted: complete case 1.38 (95% CI 1.23 to 1.55), n = 527, and multiple imputation 1.44 (95% CI 1.28 to 1.61), n = 620. Adjusted for age, sex, diabetes, chronic kidney disease and heart failure: complete case 1.39 (95% CI 1.24 to 1.57) and multiple imputation 1.45 (95% CI 1.29 to 1.64), the main analysis (Table 4, Figure 3)
5	With SOFA added: complete case 1.21 (95% CI 1.05 to 1.39), multiple imputation 1.23 (95% CI 1.07 to 1.42) (Table 4)

Q5: ICU length of stay in survivors

Block	What a correct draft contains (pack table)
1	439 survivors to day 28 with a day-3 balance: 239 with a positive and 200 with a negative balance. 366 in the adjusted model (section 2)
2	Survivors with a positive balance were sicker: median SOFA 8 (5 to 10) vs 6 (4 to 7), vasopressor 113 of 239 (47%) vs 39 of 200 (20%). Lactate missing in 73 of 439 (Table 1)
3	Median ICU stay 7.5 days (IQR 5.2 to 11.2) with a positive balance vs 4.8 (3.3 to 7.1). Mean 8.7 (SD 5.0) vs 6.0 (4.6). Geometric mean 7.4 vs 4.9 days. Range 1.2 to 27.9 vs 1.2 to 41.9 days (Table 2, Figures 1 and 2)
4	Ratio of geometric means, crude 1.50 (95% CI 1.34 to 1.68), n = 439. Adjusted 1.24 (95% CI 1.10 to 1.39), p < 0.001, n = 366 (Table 3)
5	Hodges-Lehmann difference 2.5 days (95% CI 1.8 to 3.2), Mann-Whitney p < 0.001. Per 10 ml/kg, adjusted ratio 1.06 (95% CI 1.03 to 1.08) (Table 3)

Every table and figure in the pack

Question	Tables and figures the submitted Results must include
Q1	Tables 1 to 6, Figures 1 to 4
Q2	Tables 1 to 4, Figures 1 and 2
Q3	Tables 1 to 3, Figures 1 and 2
Q4	Tables 1 to 4, Figures 1 to 3
Q5	Tables 1 to 3, Figures 1 and 2

Instructor note: running the workshop. Show the five Q1 model sentences in two or three minutes, then let everyone write on their own question for the rest of the 45 minutes. If the platform allows, put participants with the same question in one breakout room. Each still writes their own text. Visit each room and check block 1 and block 4 first, where most errors happen.

Instructor note: numbers that look inconsistent. Q2 pack section 4 gives AKI in 302 of all 620 patients. The group comparison uses the 618 with a day-3 balance (230 + 71 = 301). Q4 counts lactate as missing in 93 of 620, while Q1 to Q3 give 91 of 618 (two of the 93 have no day-3 balance). Q5 starts from 441 survivors (Q4 Table 1), of whom 439 have a day-3 balance.

Instructor note: coaching checklist for any participant's draft

Check	Typical fix
Results start with participants and flow	Add "Of 620 enrolled, 618 … 527 …" (pack section 2)
Crude AND adjusted estimates (STROBE 16)	Add the missing one, with n
Every table and figure from the pack included and cited	Add the missing ones (see the list above). Results left out are the most common gap
No interpretation in Results	Move "because sicker patients…" to the Discussion
Every effect has a 95% CI	Copy it from the pack

Check	Typical fix
Q4: complete-case AND imputed	Q4 pack Table 4: CC OR 1.39 (1.24 to 1.57), n = 527, vs MI 1.45 (1.29 to 1.64), n = 620, per mmol/L, adjusted for age, sex, diabetes, CKD and heart failure
Q5: medians and IQR, survivors only	Q5 pack section 2: 439 survivors. Q5 pack Table 2: median ICU stay 7.5 (5.2 to 11.2) vs 4.8 (3.3 to 7.1) days
Tables stand alone	Title, units, n per column, abbreviations in a footnote
Working title and AI-use statement present	Add both, e.g. "AI use: none." or the one-line statement from the brief
Methods (if submitted): a method for every result	Add the test or model and the covariates

Instructor note: model Methods for Q1 (optional section, ~300 words)

Study design and setting. We analysed FLUID-ICU, a multicentre prospective cohort of consecutive adults with sepsis admitted to four intensive care units (ICUs A–D) between January 2023 and December 2024. FLUID-ICU is a simulated teaching dataset. No ethics approval or consent was required.

Participants. All adults with sepsis enrolled in the cohort were eligible. Patients without a cumulative fluid balance on day 3 were excluded from the exposure analysis.

Variables. The exposure was a positive cumulative fluid balance by ICU day 3, defined as total fluid input minus total fluid output over ICU days 1–3 greater than 0 ml/kg, compared with a negative or zero balance. The primary outcome was death from any cause within 28 days of ICU admission. For survival analysis, time to death was censored at day 28. Confounders were pre-specified in the statistical analysis plan: age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease. Heart failure was pre-specified as a possible effect modifier.

Data sources and bias. Data were collected from ICU charts on a standard case report form. To address confounding by indication (sicker patients receive more fluid), we adjusted for markers of illness severity and comorbidity.

Study size. All eligible patients in the cohort were included. No separate sample-size calculation was performed.

Statistical analysis. Baseline characteristics are presented as mean (SD), median (IQR) or n (%), by fluid-balance group. 28-day mortality was compared with the chi-square test, and crude and adjusted odds ratios (ORs) were estimated by logistic regression. Survival to day 28 was described with Kaplan–Meier curves (log-rank test) and analysed with Cox regression using the same covariates. The proportional-hazards assumption was checked. Effect modification by heart failure was tested with an interaction term. The main analysis used complete cases. Missing lactate was handled by multiple imputation (20 data sets) in a sensitivity analysis. Estimates are reported with 95% confidence intervals (CIs). Analyses used R 4.6 (packages gtsummary, survival, mice).